Metric Markov Cotype Two of ℓ_1

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October 5, 2026

Abstract

We prove that the real Banach space ℓ_1 has metric Markov cotype two, answering a question of Mendel and Naor. As a consequence, every Lipschitz map from an arbitrary subset of a real Hilbert space into ℓ_1 extends to the whole Hilbert space with a universal multiplicative loss in its Lipschitz constant, resolving Ball's extension problem for this target.

1 Introduction

Metric Markov cotype asks whether the values of a map on a finite reversible Markov chain can be modified so that their one-step variation becomes small, while the cost of the modification is controlled by the original variation over a longer time interval. The modified values are not prescribed linear averages. This freedom is essential for the result proved here.

We work first with the real Banach space

$$\ell_1 = \left\{ x = (x(k))_{k\ge 1} : \sum_{k\ge 1} |x(k)| < \infty \right\}, \qquad \|x\|_1 = \sum_{k\ge 1} |x(k)|.$$

Write $\mathbb{N} = \{1, 2, \dots\}$ and $[n] = \{1, \dots, n\}$. A stochastic matrix $A = (a_{ij})_{i,j\in[n]}$ has nonnegative entries and each row sum equal to one. It is reversible with respect to a probability vector $\pi = (\pi_i)_{i\in[n]}$ if $\pi_i a_{ij} = \pi_j a_{ji}$ for all i, j. Reversibility implies stationarity, $\pi A = \pi$. Zero coordinates of π are allowed.

Definition 1.1 (Mendel–Naor)**.** A metric space (X, d) has metric Markov cotype two with constant $C > 0$ if, for every $n, t \in \mathbb{N}$, every probability vector π, every stochastic matrix A reversible with respect to π, and every $x_1, \dots, x_n \in X$, there exist $y_1, \dots, y_n \in X$ such that

$$\sum_i \pi_i d(x_i, y_i)^2 + t \sum_{i,j} \pi_i a_{ij} d(y_i, y_j)^2 \le C^2 \sum_{i,j} \pi_i \left(\frac{1}{t} \sum_{s=1}^{t} A^s \right)_{ij} d(x_i, x_j)^2. \tag{1.1}$$

The infimum of all such constants is denoted $N_2(X)$.

Remark (notation). X denotes the metric space whose metric Markov cotype or point configuration is under consideration. In Theorem 1.2: $X = \ell_1$ over the real scalars, equipped with its norm metric. In Corollary 6.1: The space whose cotype is asserted is $X = \ell_1(\mathbb{C})$ with its norm metric; the application of the real theorem uses $\ell_1(\mathbb{R})$. $C > 0$ is an admissible metric Markov cotype two constant for (X, d): the same C works in (1.1) for every $n, t \in \mathbb{N}$, probability vector π, stochastic matrix A reversible with respect to π, and choice $x_1, \dots, x_n \in X$, with the points

$y_1, \dots, y_n \in X$ allowed to depend on these data. In Theorem 1.2: For $X = \ell_1$, the proof takes $C^2 = 3024$, equivalently $C = 12\sqrt{21}$. The points x_i, indexed by $i \in [n]$, are the original input points in the ambient metric space. In Lemma 2.1, Theorem 1.2: The points satisfy $x_i \in \ell_1(\mathbb{R})$. In Corollary 6.1: The points satisfy $x_i \in \ell_1(\mathbb{C})$; their images Jx_i are the inputs to the real-space construction. For $i \in [n]$, $y_i \in X$ denotes a choice of modified point in the metric Markov cotype inequality. In Theorem 1.2: If all x_i coincide, choose $y_i = x_i$; otherwise choose $y_i = T(\Phi(h_i))$, where $h_i = p\sum_{s=0}^{\infty} q^s (A^s z)_i$, $p = 1/(t+1)$ and $q = 1-p$. In Corollary 6.1: For complex input points, $y_i = J^{-1}\widetilde{y}_i$, where $\widetilde{y}_i \in \ell_1(\mathbb{R})$ are the modified points obtained by applying the real construction to Jx_i.

This is [4, Definition 1.2]. Our main result is the following.

Theorem 1.2. *The real space ℓ_1 satisfies*

$$N_2(\ell_1) \le 12\sqrt{21}.$$

Remark. In the cost identities, $B = \sum_i \pi_i \|z_i - \Phi(h_i)\|_{1,w}^2$; occurrences of $B \in \mathcal{B}$ in coordinate positions denote cut indices.

Thus the square of the constant in (1.1) may be taken to be 3024, independently of the number of points, the number of coordinates, and the time parameter. No optimization of this numerical constant is intended.

History and significance

Ball introduced Markov type and cotype in his study of Lipschitz extension [1]. His Markov cotype condition for Banach spaces uses specified linear averages, and ℓ_1 fails that stronger condition. Ball also proposed a metric definition. Mendel and Naor developed the Cesàro formulation above, proved its equivalence to Ball's metric definition, and asked explicitly whether $N_2(\ell_1)$ is finite [4, Question 1.15 and Section 7]. Theorem 1.2 answers that question affirmatively.

For $1 < p \le 2$, the uniformly convex theory gives $N_2(\ell_p) \le C/\sqrt{p-1}$ with a universal constant C [1]; see also [4, Section 1.1]. These bounds deteriorate as p decreases to one and do not settle the endpoint treated here.

Remark (notation). p is a real parameter whose exponent and geometric-probability uses are local to their respective contexts. In Lemma 4.1, Theorem 1.2: $p = 1/(t+1)$ for $t \in \mathbb{N}$, so $0 < p \le 1/2$; it is the stopping probability and $q = 1-p$.

There is a second distinction worth making. Using a construction of Kalton, Mendel and Naor found a closed linear subspace of ℓ_1 that fails metric Markov cotype at every finite exponent [4, Theorem 1.14]. This does not rule out Theorem 1.2, because the points y_i in Definition 1.1 need not belong to a given subspace containing the original data. Our construction uses this freedom.

One important consequence concerns extension of Lipschitz maps. For a map f between metric spaces, $\mathrm{Lip}(f)$ denotes the least $L \ge 0$ such that $d(f(x), f(y)) \le L d(x, y)$ for all x, y in its domain. Ball's extension problem asks whether a Lipschitz map from an arbitrary subset of Hilbert space into ℓ_1 admits an extension with a universal multiplicative loss in its Lipschitz constant. Mendel and Naor explain that an affirmative answer to their metric Markov cotype question gives such an extension [4, Corollary 1.13 and the discussion after Question 1.15]. The extension problem is also discussed in [5, Section 1]. We record the consequence precisely in Section 6; the proof of Theorem 1.2 itself is elementary and independent of extension theorems.

Remark (notation). f denotes a map between the domain and target metric spaces specified in its context. In Corollary 6.2: $f : S \to \ell_1$ is Lipschitz, where $S \subseteq \mathcal{H}$ and ℓ_1 is the real sequence space.

The nonlinear construction

Finite subsets of ℓ_1 admit exact representations by weighted cuts; see [3]. We retain not only their distances but also a contractive map back to the given ambient ℓ_1. Nonlinear cut smoothing with the cubic $\phi(r) = 3r^2 - 2r^3$ also appears in the work of Cheng, Wang, and Xiang on sharp metric cotype [2]; metric cotype and metric Markov cotype are different invariants.

The additional ingredient here is a martingale estimate for the cubic. For a finite family $\mathcal{B}$ of cuts, the representation gives binary vectors $z_i \in \{0,1\}^{\mathcal{B}}$ in a weighted Hilbert space H, with

$$\|z_i - z_j\|_H^2 = \|x_i - x_j\|_1.$$

It also gives a map T with $T(z_i) = x_i$ that is contractive for the weighted ℓ_1 norm on the cut coordinates. Let $Q = [0,1]^{\mathcal{B}}$, and let $\Phi : Q \to Q$ apply ϕ coordinatewise. Stop an A-walk (X_s) after an independent nonnegative geometric number S of steps with mean t. Writing $\mathbb{E}_i$ for expectation when the walk starts at i, we choose

$$h_i = \mathbb{E}_i z_{X_S}, \qquad y_i = T(\Phi(h_i)).$$

Remark (notation). $\mathcal{B}$ is a finite nonempty coordinate index set. In Lemma 2.1: $\mathcal{B} = \{B \subseteq [n] : B \neq \varnothing,\ B \neq [n],\ \|b_B\|_1 > 0\}$ in the cut construction for the noncoincident input points. X_s is the state at integer time $s \geq 0$ of the stationary Markov walk on $[n]$. In Theorem 1.2: The walk has transition matrix A and stationary distribution π; X_s indexes the underlying walk after s live transitions, so the index at death is X_S. S is local notation for a geometric number of walk steps or for a subset of a Hilbert space, according to context. In Lemma 4.1, Theorem 1.2: $S \in \{0,1,2,\ldots\}$ is independent of the walk and satisfies $\mathbb{P}(S = s) = pq^s$, where $p = 1/(t+1)$ and $q = 1 - p$; it counts live walk transitions before stopping and has mean t. In Corollary 6.2: S is an arbitrary subset of $\mathcal{H}$, serving as the domain of $f : S \to \ell_1$. $\mathbb{E}_i$ denotes expectation under the law of an A-walk with $X_0 = i$, including the independent variable S with $\mathbb{P}(S = s) = pq^s$ for $s \geq 0$ when the integrand involves S.

For the estimate, start the walk from its stationary law π and record stopping as a separate transition after the S walk steps. Attach h_j to the current index j while alive and z_j when the walk stops at j, keeping that value thereafter. These values form a martingale. Because ϕ' vanishes at the binary values, a fourth-power Hilbert potential controls the total squared weighted-ℓ_1 variation after applying Φ. Centering at z_{X_0} makes the terminal potential equal to $\|x_{X_S} - x_{X_0}\|_1^2$. The stopping jump accounts for the approximation cost, and the expected t preceding jumps account for the one-step variation. Contractivity of T transfers these bounds to ℓ_1.

We prove the cut representation in Section 2 and the martingale estimate in Section 3. The comparison in Section 4, valid in any metric space, converts the terminal geometric endpoint cost to the Cesàro average in (1.1). These ingredients are assembled in Section 5. The resulting points have a simple interpretation: each y_i is the expected coordinatewise median of three independent geometric-walk endpoints started at i. This interpretation is established after the main proof.

2 Finite cuts and reconstruction

We first replace finitely many vectors in ℓ_1 by finitely many binary coordinates. The reconstruction map will allow us to choose new points in the full ambient space, rather than in the span of the data. This is the cut representation [3, Chapter 4], together with the ambient reconstruction given below.

Lemma 2.1. *Given $x_1, \dots, x_n \in \ell_1$, either all the points coincide, or there exist a finite set $\mathcal{B}$, positive weights $(w_B)_{B\in\mathcal{B}}$, binary vectors $z_i \in \{0,1\}^{\mathcal{B}}$, and an affine map $T : \mathbb{R}^{\mathcal{B}} \to \ell_1$ with the following properties. For*

$$\|u\|_{1,w} = \sum_{B\in\mathcal{B}} w_B |u_B|, \qquad \|u\|_H^2 = \sum_{B\in\mathcal{B}} w_B u_B^2,$$

one has

$$T(z_i) = x_i, \tag{2.1}$$

$$\|x_i - x_j\|_1 = \|z_i - z_j\|_{1,w} = \|z_i - z_j\|_H^2, \tag{2.2}$$

$$\|T(u) - T(u')\|_1 \le \|u - u'\|_{1,w} \qquad (u, u' \in \mathbb{R}^{\mathcal{B}}). \tag{2.3}$$

The space H is $\mathbb{R}^{\mathcal{B}}$ with the indicated Hilbert norm.

Remark (notation). The notation B is overloaded between a coordinate index and the nonnegative scalar approximation cost $\sum_i \pi_i \|z_i - \Phi(h_i)\|_{1,w}^2$. In Lemma 2.1: In the cut construction, B ranges over nonempty proper subsets of $[n]$; in coordinate sums, $B \in \mathcal{B}$, where only subsets with $w_B > 0$ are retained. In Lemma 3.1, Corollary 3.2: The symbol $B \in \mathcal{B}$ is a coordinate index in an arbitrary finite nonempty set with positive weights; no realization as a subset of $[n]$ is required. In Theorem 1.2: In coordinate subscripts and sums, $B \in \mathcal{B}$ is a retained cut index. In the cost identities and inequalities, the standalone symbol denotes $B = \sum_i \pi_i \|z_i - \Phi(h_i)\|_{1,w}^2$. $w_B > 0$ for every $B \in \mathcal{B}$. In Lemma 2.1: $w_B = \|b_B\|_1$ for each retained cut $B \in \mathcal{B}$; cuts with zero weight are discarded.

Proof. For each original coordinate k, put $v(k) = \min_i x_i(k)$. For every nonempty proper subset $B \subset [n]$, define the nonnegative vector

$$b_B(k) = \left(\min_{i\in B} x_i(k) - \max_{i\notin B} x_i(k) \right)_+ . \tag{2.4}$$

Here $r_+ = \max\{r, 0\}$. The estimates

$$|v(k)| \le \sum_i |x_i(k)|, \qquad b_B(k) \le 2 \sum_i |x_i(k)|$$

show that $v, b_B \in \ell_1$. Set $w_B = \|b_B\|_1$ and retain only subsets with $w_B > 0$; these form $\mathcal{B}$. There are at most $2^n - 2$ of them. Define

$$z_i(B) = \mathbf{1}_{\{i\in B\}}, \qquad T(u) = v + \sum_{B\in\mathcal{B}} u_B b_B. \tag{2.5}$$

The sum is finite, so T takes values in ℓ_1.

To verify the identities, fix k and list the distinct values among $x_1(k), \dots, x_n(k)$ as $\alpha_1 < \cdots < \alpha_m$. For $1 \le r < m$, the upper cut

$$B_r = \{i : x_i(k) \ge \alpha_{r+1}\}$$

has $b_{B_r}(k) = \alpha_{r+1} - \alpha_r$; all other cuts have $b_B(k) = 0$. Thus $T(z_i)(k)$ is α_1 plus exactly the successive gaps below $x_i(k)$, proving (2.1). Moreover, the cuts with positive $b_B(k)$ are nested.

If $x_i(k) \geq x_j(k)$, every nonzero term in $\sum_B (z_i(B) - z_j(B))b_B(k)$ is nonnegative; the reverse inequality gives the reverse sign. Hence

$$|x_i(k) - x_j(k)| = \sum_B |z_i(B) - z_j(B)| b_B(k).$$

Sum over k. Since binary differences have equal absolute value and square, this gives (2.2). The triangle inequality applied to (2.5) gives (2.3). If $\mathcal{B}$ is empty, the same coordinate description shows that all the data coincide. □

3 A flat map and a quartic potential

By (2.2), squared ℓ_1 distances between the original data are fourth powers of Hilbert distances between their binary encodings. We therefore seek a fourth-power potential that controls squared increments after the cubic map.

Fix a finite nonempty set $\mathcal{B}$ and positive weights w_B, and use the two norms from Lemma 2.1. Write $Q = [0,1]^{\mathcal{B}}$ and let $\langle \cdot, \cdot \rangle_H$ be the weighted Hilbert inner product. Define

$$\phi(r) = 3r^2 - 2r^3 \quad (0 \leq r \leq 1), \qquad \Phi(u)_B = \phi(u_B) \quad (u \in Q).$$

The map Φ takes Q to itself and fixes its binary vertices. The key estimate controls its squared weighted-ℓ_1 increment by a convexity remainder in H.

Lemma 3.1. *For $u, u' \in Q$ and $e \in \{0,1\}^{\mathcal{B}}$, put $a = u - e$, $\delta = u' - u$, and $F_e(u) = \|u - e\|_H^4$. Then*

$$\|\Phi(u') - \Phi(u)\|_{1,w}^2 \leq 108\big(F_e(u') - F_e(u) - 4\|a\|_H^2 \langle a, \delta \rangle_H\big). \tag{3.1}$$

Remark (notation). e is a binary vector in $\{0,1\}^{\mathcal{B}}$ used as the center of the quartic potential. In Lemma 3.1: e is an arbitrary fixed binary vector. In Corollary 3.2: e is $\mathcal{F}_0$-measurable. In Theorem 1.2: In the stopped-walk construction, $e = z_{X_0}$ and is $\mathcal{F}_0$-measurable.

Proof. For either bit $d \in \{0,1\}$ and $r \in [0,1]$,

$$|\phi'(r)| = 6r(1-r) \leq 6|r - d|.$$

Integrating on the segment from u_B to u'_B gives

$$\begin{aligned} |\phi(u'_B) - \phi(u_B)| &\leq 6|\delta_B| \int_0^1 |a_B + s\delta_B| \, ds \\ &\leq 6|a_B||\delta_B| + 3|\delta_B|^2. \end{aligned}$$

Summing with weights and applying Cauchy–Schwarz yields

$$\|\Phi(u') - \Phi(u)\|_{1,w} \leq 6\|a\|_H \|\delta\|_H + 3\|\delta\|_H^2. \tag{3.2}$$

Let R denote the expression in parentheses in (3.1). Expanding the fourth power gives the exact identity

$$R = 2\|a\|_H^2 \|\delta\|_H^2 + \big(2\langle a, \delta \rangle_H + \|\delta\|_H^2\big)^2. \tag{3.3}$$

In particular, $\|a\|_H^2\|\delta\|_H^2 \le R/2$. Also

$$\begin{aligned}\|\delta\|_H^4 &\le 2\big(2\langle a,\delta\rangle_H + \|\delta\|_H^2\big)^2 + 8\langle a,\delta\rangle_H^2 \\ &\le 2\big(2\langle a,\delta\rangle_H + \|\delta\|_H^2\big)^2 + 8\|a\|_H^2\|\delta\|_H^2 \le 4R.\end{aligned}$$

Squaring (3.2) and using $(r+s)^2 \le 2r^2 + 2s^2$ therefore gives

$$\|\Phi(u') - \Phi(u)\|_{1,w}^2 \le 72\|a\|_H^2\|\delta\|_H^2 + 18\|\delta\|_H^4 \le 36R + 72R = 108R.$$

□

Remark (notation). R denotes a nonnegative remainder, with its definition local to its occurrence. In Lemma 3.1: $R = F_e(u') - F_e(u) - 4\|a\|_H^2\langle a,\delta\rangle_H = 2\|a\|_H^2\|\delta\|_H^2 + (2\langle a,\delta\rangle_H + \|\delta\|_H^2)^2$. In Lemma 4.1: In the proof, $R = S - t\lfloor S/t\rfloor$ takes values in $\{0,\dots,t-1\}$, and $\mathbb{P}(R=r) = pq^r/(1-q^t)$ for $0 \le r < t$.

The linear term in (3.1) disappears on taking a conditional expectation along a Hilbert-valued martingale. It is important that the center may depend on the initial state.

Corollary 3.2. *Let $(M_m)_{m\ge 0}$ be an H-valued martingale with respect to a filtration $(\mathcal{F}_m)$, taking values in Q and converging almost surely to M_∞. If e is an $\mathcal{F}_0$-measurable binary vector, then*

$$\begin{aligned}\mathbb{E}\sum_{m=0}^{\infty}\|\Phi(M_{m+1}) - \Phi(M_m)\|_{1,w}^2 &\le 108\big(\mathbb{E}\|M_\infty - e\|_H^4 - \mathbb{E}\|M_0 - e\|_H^4\big) \\ &\le 108\mathbb{E}\|M_\infty - e\|_H^4. \qquad (3.4)\end{aligned}$$

Remark (notation). $(M_m)_{m\ge 0}$ is an H-valued martingale with respect to $(\mathcal{F}_m)$, taking values in Q and converging almost surely to M_∞. In Theorem 1.2: M_m is the value attached to the state at chain time m: the value is h_i at an alive state of index i and z_i at a dead state of index i; in particular, $M_0 = h_{X_0}$ and $M_m = z_{X_S}$ for $m \ge S+1$. $\mathcal{F}_m$, for integers $m \ge 0$, is the sigma algebra at time m of the filtration with respect to which $(M_m)_{m\ge 0}$ is a martingale. In Theorem 1.2: $\mathcal{F}_m$ is generated by the initial state and the first m transitions of the chain on alive and dead states, with dead states absorbing; it records no future transitions or survival trials. $M_\infty = \lim_{m\to\infty} M_m$ almost surely, with values in Q. In Theorem 1.2: $M_\infty = z_{X_S}$. $M > 0$ is a Markov type two constant for the given metric space (Z, d_Z), uniform over every $n, t \in \mathbb{N}$, every probability vector π, every stochastic matrix A reversible with respect to π, and all $u_1,\dots,u_n \in Z$. In Corollary 6.2: For the Hilbert source $\mathcal{H}$, the Markov type two constant is taken to be $M = 1$.

Proof. Apply Lemma 3.1 to each pair M_m, M_{m+1}. The vector $4\|M_m - e\|_H^2(M_m - e)$ is $\mathcal{F}_m$-measurable and bounded. Therefore

$$\mathbb{E}\big[\|M_m - e\|_H^2\langle M_m - e, M_{m+1} - M_m\rangle_H \mid \mathcal{F}_m\big] = 0.$$

Taking expectations and summing over $0 \le m < L$ telescopes the potentials. All potentials are bounded by $(\sum_B w_B)^2$, so bounded convergence applies to the terminal potential as $L \to \infty$. Monotone convergence applies to the nonnegative jump sum. This proves the first inequality; the second follows by discarding the initial nonnegative potential. □

4 Geometric and Cesàro endpoint costs

The stopped martingale will produce a geometric average of endpoint distances. The next estimate converts it to the finite average in Definition 1.1. This comparison is part of the resolvent approach in [4, Section 7, especially Lemma 7.1]. We give an elementary version with an explicit constant; only stationarity, not reversibility, is needed.

Lemma 4.1. *Let $(X_s)_{s\ge 0}$ be a stationary Markov chain on $[n]$, and let $x_1,\dots,x_n$ lie in a metric space (X,d). For $t\in\mathbb{N}$, set $p=1/(t+1)$ and $q=1-p$. If S is independent of the walk and $\mathbb{P}(S=s)=pq^s$ for $s\ge 0$, then*

$$\mathbb{E}d(x_{X_S},x_{X_0})^2 \le \frac{28}{t}\sum_{s=1}^{t}\mathbb{E}d(x_{X_s},x_{X_0})^2. \tag{4.1}$$

Proof. Write

$$D(s)=\mathbb{E}d(x_{X_s},x_{X_0})^2, \qquad W=\frac{1}{t}\sum_{s=1}^{t}D(s).$$

The triangle inequality, Minkowski's inequality in L_2, and stationarity imply

$$\sqrt{D(r+s)}\le\sqrt{D(r)}+\sqrt{D(s)} \qquad (r,s\ge 0). \tag{4.2}$$

Consequently $D(t)\le 2D(r)+2D(t-r)$ for $1\le r\le t$. Since $D(0)=0$, averaging gives

$$D(t)\le\frac{2}{t}\sum_{r=1}^{t}D(r)+\frac{2}{t}\sum_{r=0}^{t-1}D(r)\le 4W. \tag{4.3}$$

If $s=jt+r$ with $j\ge 0$ and $0\le r<t$, repeated use of (4.2) gives

$$D(s)\le\left(j\sqrt{D(t)}+\sqrt{D(r)}\right)^2\le 2j^2D(t)+2D(r). \tag{4.4}$$

For $J=\lfloor S/t\rfloor$ and $R=S-tJ$, the geometric second moment yields

$$\mathbb{E}J^2\le\frac{\mathbb{E}S^2}{t^2}=\frac{2t^2+t}{t^2}\le 3.$$

For each $0\le r<t$,

$$\mathbb{P}(R=r)=\frac{pq^r}{1-q^t}\le\frac{2}{t}.$$

Indeed $(1+1/t)^t\ge 2$ implies $q^t\le 1/2$, so the left side is at most $2p=2/(t+1)\le 2/t$. Thus

$$\mathbb{E}D(R)\le\frac{2}{t}\sum_{r=0}^{t-1}D(r)\le 2W.$$

Take expectations in (4.4) and use independence of S and the walk to conclude

$$\mathbb{E}d(x_{X_S},x_{X_0})^2=\mathbb{E}D(S)\le 2\cdot 3\cdot 4W+2\cdot 2W=28W.$$

Every step remains valid when $t=1$; then $R=0$. □

Remark (notation). The notation J is local to the proof in which it occurs. In Lemma 4.1: $J=\lfloor S/t\rfloor$ is a nonnegative integer-valued random quantity defined from the geometric stopping time, and $R=S-tJ$. In Corollary 6.1: $J:\ell_1(\mathbb{C})\to\ell_1(\mathbb{R})$ is the real-linear bijection $Jx=(\Re x(1),\Im x(1),\Re x(2),\Im x(2),\dots)$, satisfying $\|x\|_1\le\|Jx\|_1\le\sqrt{2}\|x\|_1$.

5 The stopped walk and the main theorem

We now construct the required points. Fix the data in Definition 1.1 with $X = \ell_1$. If all x_i coincide, choose $y_i = x_i$. Otherwise use Lemma 2.1 to obtain z_i, H, T, and use the map Φ of Section 3.

Put $p = 1/(t+1)$, $q = 1-p$, and define

$$h_i = p\sum_{s=0}^{\infty} q^s (A^s z)_i, \qquad (A^s z)_i = \sum_j (A^s)_{ij} z_j. \tag{5.1}$$

This convex combination converges in the finite-dimensional space H and belongs to Q. Equivalently, $h_i = \mathbb{E}_i z_{X_S}$, where $\mathbb{E}_i$ refers to an A-walk started at i and S is independent with $\mathbb{P}(S = s) = pq^s$. The first-step identity is

$$h_i = pz_i + q\sum_j a_{ij} h_j. \tag{5.2}$$

We take

$$y_i = T(\Phi(h_i)). \tag{5.3}$$

In particular, each y_i belongs to ℓ_1.

5.1 A martingale that realizes the two costs

Start a chain in an alive state with index distributed according to π. From an alive state of index i, it moves to the dead state of index i with probability p, and to the alive state of index j with probability qa_{ij}. Dead states are absorbing. Its filtration $\mathcal{F}_m$ records the initial state and the first m transitions only.

One realization uses an independent stationary A-walk $(X_s)_{s\ge 0}$ and successive independent survival trials, each successful with probability q. Let S be the number of successful trials before the first failure. Then $\mathbb{P}(S = s) = pq^s$ and S is independent of the walk. There are S live index transitions followed by one death transition; the index at death is X_S.

Attach the value h_i to the alive state of index i and the value z_i to its dead state, and let M_m be the attached value at chain time m. Equation (5.2) shows that (M_m) is an H-valued martingale for the indicated filtration. It lies in Q and is eventually constant almost surely, with

$$M_0 = h_{X_0}, \qquad M_\infty = z_{X_S}.$$

The vector $e = z_{X_0}$ is $\mathcal{F}_0$-measurable and binary. Corollary 3.2 and (2.2) give

$$\mathbb{E}\sum_{m=0}^{\infty} \|\Phi(M_{m+1}) - \Phi(M_m)\|_{1,w}^2 \le 108\mathbb{E}\|x_{X_S} - x_{X_0}\|_1^2. \tag{5.4}$$

To compute the left side exactly, set

$$B = \sum_i \pi_i \|z_i - \Phi(h_i)\|_{1,w}^2, \qquad E_A = \sum_{i,j} \pi_i a_{ij} \|\Phi(h_i) - \Phi(h_j)\|_{1,w}^2.$$

At time m, the event that the chain is alive has probability q^m and is independent of X_m. Stationarity gives $\mathbb{P}(X_m = i) = \pi_i$ and $\mathbb{P}(X_m = i, X_{m+1} = j) = \pi_i a_{ij}$. Since $\Phi(z_i) = z_i$, the expected squared jump at time m is $pq^m B + q^{m+1} E_A$. Therefore

$$\mathbb{E}\sum_{m=0}^{\infty} \|\Phi(M_{m+1}) - \Phi(M_m)\|_{1,w}^2 = B + \frac{q}{p} E_A = B + tE_A. \tag{5.5}$$

This identity explains why the killing probability was chosen to be $1/(t+1)$.

5.2 Completion of the proof

Proof of Theorem 1.2. Use the points (5.3). Contractivity of T and the identity $T(z_i) = x_i$ imply

$$\sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \\ \leq B + tE_A \leq 108 \mathbb{E} \|x_{X_S} - x_{X_0}\|_1^2,$$

where the last inequality follows from (5.4) and (5.5). Lemma 4.1 bounds the last expression by

$$108 \cdot 28 \cdot \frac{1}{t} \sum_{s=1}^{t} \mathbb{E} \|x_{X_s} - x_{X_0}\|_1^2.$$

The stationary walk satisfies

$$\frac{1}{t} \sum_{s=1}^{t} \mathbb{E} \|x_{X_s} - x_{X_0}\|_1^2 = \sum_{i,j} \pi_i \left(\frac{1}{t} \sum_{s=1}^{t} A^s \right)_{ij} \|x_i - x_j\|_1^2.$$

This is (1.1) with $C^2 = 3024$. □

Remark 5.1. The proof uses reversibility only through $\pi A = \pi$. Consequently, the same inequality and the same construction hold for every stationary stochastic matrix, without detailed balance. No positivity assumption on the individual π_i is needed in the argument.

5.3 An expected-median interpretation

The construction has an intrinsic description in the original coordinates. Let

$$G = p \sum_{s=0}^{\infty} q^s A^s$$

be the stochastic matrix of geometric endpoints. For fixed i, take independent indices J_1, J_2, J_3 with common law $\mathbb{P}(J_a = j) = G_{ij}$. For three real vectors, write $\operatorname{med}(x, x', x'')$ for their coordinatewise median. This vector belongs to ℓ_1, since its coordinatewise absolute value is at most $|x| + |x'| + |x''|$. Then

$$y_i = \mathbb{E} \operatorname{med}(x_{J_1}, x_{J_2}, x_{J_3}). \tag{5.6}$$

In each original coordinate, the median is at least a given threshold exactly when at least two of the three values are. The threshold decomposition in the proof of Lemma 2.1 therefore gives the pointwise identity

$$\operatorname{med}(x_{J_1}, x_{J_2}, x_{J_3}) = v + \sum_{B \in \mathcal{B}} b_B \mathbf{1}_{\{\#\{a \in [3] : J_a \in B\} \geq 2\}}.$$

Each indicator has expectation $3h_i(B)^2(1 - h_i(B)) + h_i(B)^3 = \phi(h_i(B))$. Taking expectations in the finite sum proves (5.6).

6 Complex scalars and Lipschitz extension

Corollary 6.1. *For complex scalars,* $N_2(\ell_1(\mathbb{C})) \le 12\sqrt{42}$.

Proof. The real-linear bijection

$$J : \ell_1(\mathbb{C}) \longrightarrow \ell_1(\mathbb{R}), \qquad Jx = (\Re x(1), \Im x(1), \Re x(2), \Im x(2), \ldots)$$

satisfies $\|x\|_1 \le \|Jx\|_1 \le \sqrt{2}\|x\|_1$. Apply Theorem 1.2 to Jx_i and pull the resulting points back by J^{-1}. The left side of (1.1) can only decrease, while its right side increases by at most a factor of 2. □

For the extension consequence, recall that a metric space (Z, d_Z) has *Markov type two* with constant $M > 0$ if

$$\sum_{i,j} \pi_i (A^t)_{ij} d_Z(u_i, u_j)^2 \le M^2 t \sum_{i,j} \pi_i a_{ij} d_Z(u_i, u_j)^2$$

for every $n, t \in \mathbb{N}$, every stochastic matrix A reversible with respect to a probability vector π, and every $u_1, \ldots, u_n \in Z$. Hilbert spaces satisfy this inequality with $M = 1$ [1]; see also [4, Definition 1.1 and the following discussion].

Corollary 6.2. *There is a universal constant* $K < \infty$ *such that, for every real Hilbert space* $\mathcal{H}$*, every subset* $S \subseteq \mathcal{H}$*, and every Lipschitz map* $f : S \to \ell_1$*, there is an extension* $F : \mathcal{H} \to \ell_1$ *with* $\mathrm{Lip}(F) \le K \,\mathrm{Lip}(f)$.

Remark. $\mathrm{Lip}(f) = \inf\{L \ge 0 : d(f(x), f(y)) \le L d(x, y)$ for all x, y in the domain of $f\}$, with $\inf \varnothing = \infty$.

Proof. We use the Lipschitz extension theorem in [4, Corollary 1.13], extending Ball's theorem [1]. In its Banach-space form, a source with Markov type two and a dual Banach target with metric Markov cotype two admit extensions with a constant bounded by a universal multiple of the two corresponding constants. A Hilbert space has Markov type two with constant one, and ℓ_1 is the dual of c_0. Theorem 1.2 verifies the remaining hypothesis. □

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